

**DEPARTMENT OF MINING ENGINEERING
MID SEMESTER EXAM**

Monsoon Semester Examination: V SEM B.Tech (ME)

Subject: Surface Mining (MEC-15102)

Instruction: Answer all questions

Session: 2019-20

Time: 2 Hours

Max. Marks: 30

Sl. No.	Question	Marks
1.	a) Draw a balancing diagram for simple side cast operation of a dragline and derive the expression, in terms of known parameter, for i.height of spoil heap ii.reach of the dragline b) What is meant by MUF of a dragline and what is its physical significance?	8+2=10
2.	Estimate the number of drills required to achieve a production of 5 million ton/year of coal from a surface mine having coal seam of thickness 10 m. The coal seam is flat and horizontal and the stripping ratio of the mine is $3 m^3/ton$. (Assume relevant data wherever necessary and mention them)	(10)
3.	With the help of suitable sketch derive the expressions for optimum spacing between parallel passes in terms of known parameters in case of ripper operation on a horizontal area. OR Enlist and discuss the unit operations that are accomplished in a surface mine using discontinuous/cyclic system of mining. Also indicate the equipment to be used for each of the unit operations.	(10)